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Civil Engineering
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B.Tech.
Gender: Male
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Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2026	
Intermediate	CBSE	Tilak Public School	2022	
Matriculation	CBSE	Tilak Public School	2020	

Pursuing a **Minor** in **Data Science** offered by **C-MInDS**(Centre of Machine Intelligence and Data Science), IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Among the top **2 percentile** in JEE Advanced Examination out of **0.16 million** candidates all over India (2022)
- Achieved a position in the top **2500** among **1.1 million** applicants all over India in the JEE Main Examination (2022)

PROFESSIONAL EXPERIENCE

KlimaShift | Software Engineering & Machine Learning Intern [Jun'25-Aug'25]

An international tech startup building AI-powered smart energy solutions for small-scale commercial buildings globally

ML Ops & Backend Development	• Engineered & deployed 10+ internal APIs with FastAPI automating ML training-validation pipeline • Developed a frontend of ML platform using Dash integrating internal APIs to streamline workflow from model training to precise image annotation and bounding box drawing with Dash-Canvas • Ensured API reliability & robustness through data validation with Pydantic and testing with Pytest • Authored API specifications with Swagger for interactive documentation & validation of ML endpoints • Implemented version control for ML models & datasets using DVC with S3 for storage & reproducibility
Time-series Database & Alerting	• Developed scalable time-series data pipeline using VictoriaMetrics for data of 2k+ virtual meters • Dockerized robust, multi-tiered alerting system with vmalet & Alertmanager that uses custom Prometheus exporter to fetch dynamic thresholds from PostgreSQL for real-time user notification • Engineered advanced time-series features that identified 5+ previously undetectable event patterns

Ernst & Young | Data Science Intern [May'24-Jun'24]

Built a custom model predicting the employee attrition considering multiple factors using dataset of 40k+ EY employees

Approach	• Pre-processed and categorized a database consisting of 15+ attributes using Python and SQL • Automated extraction of employee office-home distances with a Python script and a GMaps API • Analysed data and identified trends through plotting graphs using Seaborn, Matplotlib and Plotly • Evaluated 4 ML algorithms, selecting the optimal model based on accuracy, F1-score, and recall
Results	• Built neural network model with 94% accuracy, 89 f1 score & 91 recall using Keras & TensorFlow • Deployed model as web service & developed interactive GUI using PyQT5 for individual analysis

KEY PROJECTS

TripSync | Trip Planning App | Self Project [Nov'24-Jan'25]

- Engineered a full-stack application using **MERN** stack & **React Native**, featuring secure **JWT**-based authentication
- Developed an itinerary system for daily plans and expense tracking with full **CRUD** functionality via **RESTful** API
- Integrated **Google Places API** to provide real-time location suggestions, enhancing user experience for trip planning

LiftLog | Fitness App | Self Project [May'24-Jun'24]

- Developed cross-platform app with **React Native**, **TailwindCSS** & **Expo** Router, featuring **1000+** exercises from API
- Personalized user experience with **AsyncStorage** for local data persistence, enabling users to save custom workouts
- Implemented workout logger for tracking sets/weights, visualizing user progress & strength gains via dynamic **graphs**

RAG-Powered Research Assistant | Course Project : CS626 | Prof Pushpak Bhattacharya [Sep'24-Nov'24]

- Engineered **RAG** system using **LangChain**, **Elasticsearch**, and **Llama 3.2** to enable real-time question answering on current events beyond the LLM's training cutoff date by leveraging semantic search and contextual prompting
- Implemented vector-based retrieval system achieving **800**-token chunk size & **100**-token overlap for optimal context
- Architected a **RAG pipeline** using LangChain & **LangGraph**, reducing hallucination through controlled LLM **temperature**

End-to-End Score Predication App | Winter in Data Science | Analytics Club | IIT Bombay [Dec'23-Jan'24]

- Developed a cricket score prediction app with **React Native** and **FastAPI** for real-time T20 & ODI score prediction
- Leveraged **EDA** to engineer features from **1.5M+** T20/ODI ball-by-ball entries for 15+ teams for separate models
- Trained and evaluated **5+** ML models, selecting Random Forest Regressor, achieving the lowest **MAE** of **7.8** runs

Deep Reinforcement Learning for Stock Trading | FinSearch | Finance Club | IIT Bombay [Jul'24-Aug'24]

- Built an RL-based trading agent with FinRL using **DQN** & **DDPG**, achieving **21%** annual return vs **8%** benchmark
- Engineered **15+** technical indicators to enhance the state space, improving reward convergence speed by **30%**
- Compared DQN and DDPG, with the optimized DDPG yielding **15%** higher Sharpe ratio and **20%** lower drawdown

LightGCN for Recommender Systems | Course Project : DS303 | Prof. Manjesh K Hanawal [Apr'25-May'25]

- Developed a movie recommendation system using the **LightGCN**(Graph Convolutional Network) & **self-supervised** learning, proving more efficient and **scalable** than the standard GCNs by eliminating costly feature transformations
- Modeled user-item interactions with **bipartite graph** from **100,000+** movie ratings for efficient embedding propagation
- Implemented a custom training pipeline in **PyTorch**, utilizing **BPR loss** function to optimize model performance

CNN-Based Emotion Classifier | Course Project : CE235 | Prof. Gopal R Patil [Sep'23-Nov'23]

- Developed a **Convolutional Neural network (CNN)**-based emotion detection model with **83%** validation accuracy, utilizing **Keras** and **TensorFlow**, and employing layered advanced data preprocessing and augmentation techniques
- Successfully implemented real-time emotion recognition using **Haar cascades** and **OpenCV** for live video feed

Virtual Mouse Control using Hand Gestures | Self Project [May'24-Jul'24]

- Developed using **MediaPipe** and **OpenCV** for mouse control, volume adjustment, and brightness management
- Implemented **30+** Hand gestures in Python for **Scrolling, Clicking & Volume** control at high frame rates
- Designed **multi-hand** detection and **UI** interactions with **PyAutoGUI**, enhancing user experience and performance

POSITION OF RESPONSIBILITY

Department Academic Mentor | Student Mentorship Program | IIT Bombay [Jun'24-May'25]

Assisting in the management of the official student body responsible for guidance of second-year undergraduates

- Part of a **30**-member team and a **7**-member **Web** team, selected after rigorous interviews, SOPs & peer reviews
- Guided **5+** sophomores consistently on their academics, time-management, career decisions & personal endeavours
- Actively involved in bridging the student-faculty gap and ensuring proper and timely information sharing with student

Cultural Secretary | Hostel 2 Council | IIT Bombay [Aug'23-Mar'24]

Contributor of a 2-tier Cabinet of 12 members | Council Representing 450+ Students | Budget INR 0.8Mn+

- Ensured seamless execution of hostel's Annual fest with **20+** events, institute-wide footfall, and budget of INR **0.18M**
- Contributed to a **300+** member team's historic clean sweep, winning all **12** awards at Performing Arts Festival (PAF)

TECHNICAL SKILLS

- **Programming Languages & DBMS** : Python, C++, SQL(MySQL, PostgreSQL), NoSQL(MongoDB), JS, HTML
- **Frameworks & Libraries** : React Native, Pydantic, Scikit-learn, Matplotlib, Seaborn, PyTorch, TensorFlow, Streamlit
- **Software Tools/Platforms** : Docker, DVC, Git, GitHub, MATLAB, Microsoft Office, VS Code, AutoDesk Fusion 360

RELEVANT COURSES UNDERTAKEN

AI & ML	Speech and Natural Language Processing and the Web, Statistical Machine Learning and Data Mining, Foundations of Intelligent and Learning Agents*, Introduction to Machine Learning, Artificial Intelligence & Data Science, Computer Programming and Utilization
Math	Probabilistic and Statistical Methods in Civil Eng, Numerical Methods in Civil Engineering

EXTRA-CURRICULAR ACTIVITIES

Technical	• Completed an intensive 8-week long DSA Bootcamp organized by Coding Ninjas x WnCC • Awarded the 1st runners up out of 10+ hostel teams in Algo GC representing Hostel 2
Cultural	• Awarded the 1st position out of 10+ hostel teams in Gyrations GC dance competition • Served as the Dramatics Overall Coordinator and a member of Fine Arts Team for PAF2 • Fostered vibrant hostel community, securing 1st in Sports & Technical, 2nd in Cultural GCs
Mentorship	• NEC Mentor: Mentored 20+ teams across India to achieve 1st position out of 1500+ colleges